

(but a device may not support all speeds). There's also a newer 400 MBps version.

In order to simplify applications programming, some Applications Programming Interfaces (API) were developed. The API presents a uniform interface to the scanner. This means that the application does not need to know the specific details of the scanner in order to access it directly. For example, Adobe Photoshop supports the TWAIN standard. Consequently, (in an ideal world) Photoshop can acquire an image from any scanner that also supports TWAIN.

In practice, there are often problems with an application communicating with a scanner. Either the application or the scanner manufacturer (or both) may have faults in their implementation of the API.

In addition to the API, many scanners come bundled with other software such as a scanning utility, some type of image-editing application such as Photoshop and OCR software. OCR, or optical character recognition, enables the conversion of graphical images of text into standard text that can be edited using common word-processing and text-editing software. OCR utilises an averaging algorithm to determine the character shape, then matches that shape to a corresponding letter or number.

The scanned result is a non-compressed RGB image, which can be transferred to a computer's memory. Some scanners compress and clean up the image using embedded firmware. Once on the computer, the image can be processed with a raster graphics program, such as Photoshop, and saved on a storage device (such as a hard disk).

In common use, scanned pictures are stored on a computer's hard disk, normally in image formats such as JPEG, TIFF, Bitmap, and PNG. Some scanners can also be used to capture editable text, so long as the text can be read by the computer in a discernable font. This process is called Optical Character Recognition (OCR).